Angles & 2D Shapes

Year 5 Accelerated — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

Easy: What type of angle is 35°?
Method & answer: It is between 0° and 90°, so it is acute.

Standard: Two angles on a straight line are 65° and x°. What is x?
Method & answer: Angles on a straight line add to 180°. x = 180 − 65 = 115°.

Advanced: A quadrilateral has angles of 90°, 85° and 100°. What is the fourth angle?
Method & answer: Angles in a quadrilateral add to 360°. 360 − (90+85+100) = 360 − 275 = 85°.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. What type of angle is 40°?
2. What type of angle is 120°?
3. Which of these is a reflex angle? A) 90° B) 180° C) 210° D) 170°
4. What type of angle is 200°?
5. Angles on a straight line always add up to how many degrees?
6. Angles around a point always add up to how many degrees?
7. How many sides does a pentagon have?
8. How many sides does a hexagon have?
9. Select ALL of these that are the sum of the interior angles of a triangle: A) 180° B) 90° C) 360° D) the same for every triangle E) different for every triangle
10. How many sides does an octagon have?
11. What is the sum of the interior angles of a triangle?
12. What type of triangle has all three sides equal?
13. What type of triangle has exactly two sides equal?
14. How many lines of symmetry does a square have?
15. What is the name for an angle that is exactly 180°?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Two angles on a straight line are 72° and x°. What is x?
2. Three angles around a point are 110°, 95° and x°. What is x?
3. A triangle has angles of 50° and 65°. What is the third angle? A) 55° B) 65° C) 75° D) 115°
6. In a triangle, two angles are equal at 55° each. What is the third angle?
7. Two straight lines cross, forming vertically opposite angles. If one angle is 48°, what is the angle vertically opposite it?
8. Select ALL of these that are TRUE about vertically opposite angles: A) they are always equal B) they are formed when two lines cross C)
11. A quadrilateral has three right angles. What is the size of the fourth angle?
12. Two angles on a straight line are in the ratio 2:3. What are the two angles?
13. The exterior angle of a triangle equals the sum of the two opposite interior angles. If those are 40° and 65°, what is the exterior angle?

they always add to 180° D) there are two pairs when two lines cross E) they are always acute

4. A quadrilateral has angles of 100°, 80° and 95°. What is the fourth angle?

5. Two equal angles lie on a straight line. What is the size of each angle?

9. An isosceles triangle has a base angle of 70°. What are the other two angles?

10. A right angle is split into two angles, one of which is 35°. What is the other angle?

14. A triangle has one angle of 90° and another of 40°. What is the third angle?

15. The interior angles of a regular hexagon sum to 720°. What is each interior angle?

Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A triangle has angles in the ratio 2:3:4. What is the size of each angle?

2. Two straight lines cross, forming four angles. One angle is 3 times an adjacent angle. What are all four angles?

3. A quadrilateral has angles of x°, 2x°, 3x° and 4x°. What is x, and what is the largest angle? A) x=36, largest=144° B) x=40, largest=160° C) x=36, largest=108° D) x=30, largest=120°

4. The interior angles of a pentagon sum to 540°. What is each interior angle of a regular pentagon?

5. An exterior angle of a regular polygon is 40°. How many sides does it have? (exterior angles sum to 360°)

6. Select ALL of these that are TRUE about the exterior angles of ANY polygon: A) they always sum to 360° B) they always sum to 180° C) each one is supplementary to its interior angle D) a regular polygon's exterior angles are all equal E) they can only be found for regular polygons

7. Two angles on a straight line are (2x+10)° and (3x−10)°. What is x, and what are the two angles?

8. A kite has two equal angles of 100° each. If its other two equal angles share the rest of 360°, what is each of those angles?

9. A regular polygon has an interior angle of 150°. How many sides does it have?

10. Angles a, b and c lie around a point. a = 90° and b = 2c. Given a+b+c = 360°, find b and c.

11. A triangle has an exterior angle of 130° at one vertex. If one of the other interior angles is 35°, what is the third interior angle?

12. A parallelogram has one angle of 65°. What are its other three angles?

13. In triangle ABC, angle A is twice angle B, and angle C is 20° more than angle B. Find all three angles.

14. The interior angles of a regular polygon sum to 1440°. How many sides does it have, and what is each interior angle?

15. A transversal crosses two parallel lines. One angle formed at the first line is 72°. What is the co-interior angle at the second line, on the same side?